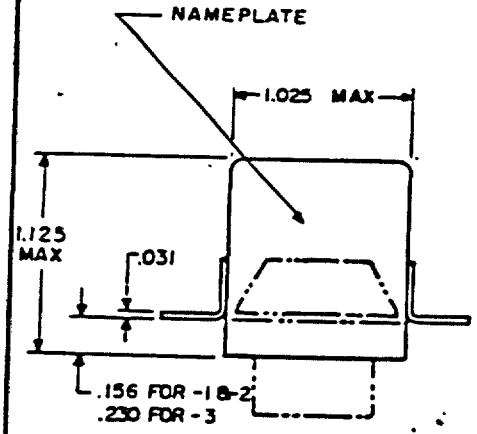
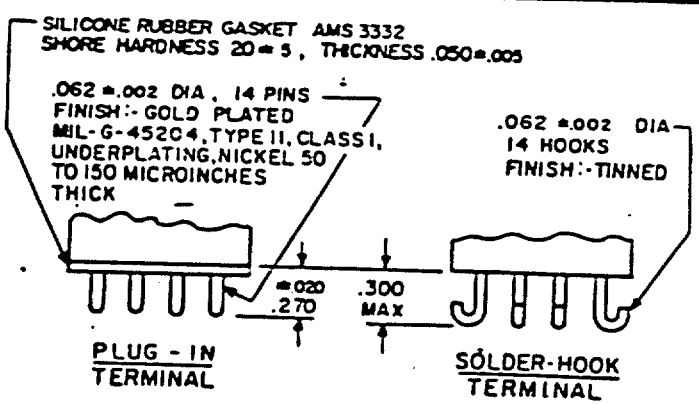
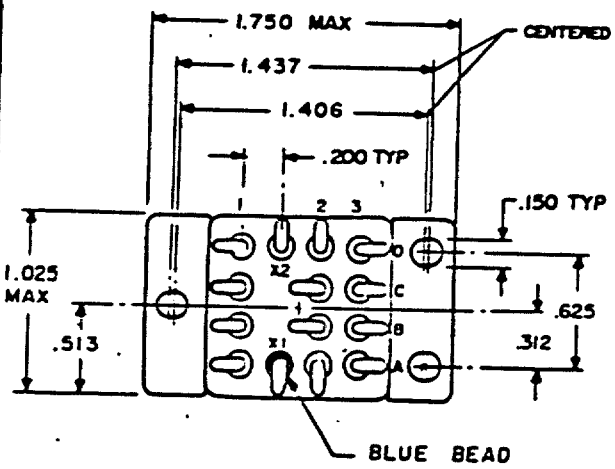
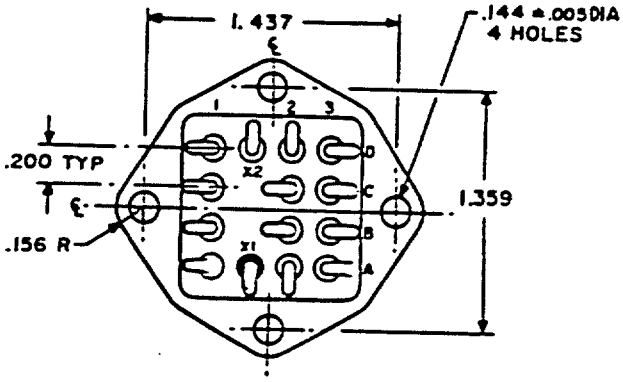
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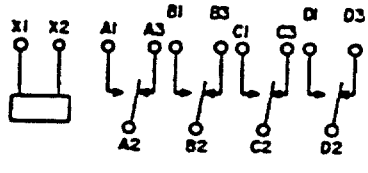
FLANGE (3 HOLE)
MS27709 -18-2



FLANGE (4 HOLE)
MS27709-3

MS PART NUMBER	TYPE	COIL TYPE	TERMINAL TYPE	(F) MOUNTING OR MATING SOCKET	MAX WEIGHT IN LBS
MS27709-1	I	DC	SOLDER HOOK	FLANGE (3 HOLE)	0.2
-2			PLUG-IN	M 12883/40,-01,-05,-07, -11,-13,-17,-19 OR -23.	0.2
-3			SOLDER HOOK	FLANGE (4 HOLE)	0.2
-4			SOLDER HOOK	STUD	0.2
-5			POTTED LEADS	STUD	0.5
-6	I	AC	SOLDED HOOK	STUD	0.5

LEAD CODING			
TERM	COLOR	TERM	COLOR
A1	BLUE	C2	WHITE-BROWN
A2	BROWN	C3	WHITE - RED
A3	RED	D1	WHITE-YELLOW
B1	YELLOW	D2	WHITE-ORANGE
B2	ORANGE	D3	WHITE-GREEN
B3	GREEN	X1	WHITE
C1	WHITE - BLUE	X2	BLACK



CIRCUIT

(F) Denotes change

P.A AF - 85 Other Cust NAVY - AS	International Interest	TITLE (F) RELAY, 10 AMPERES, 4 PDT, TYPE I HERMETICALLY SEALED	MILITARY STANDARD MS27709
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 3

DD FORM 1 MAY 73 672 (Coordinated page-block portions of this form are obsolete)

5945-0680

User activities: Army - Navy - Air Force -

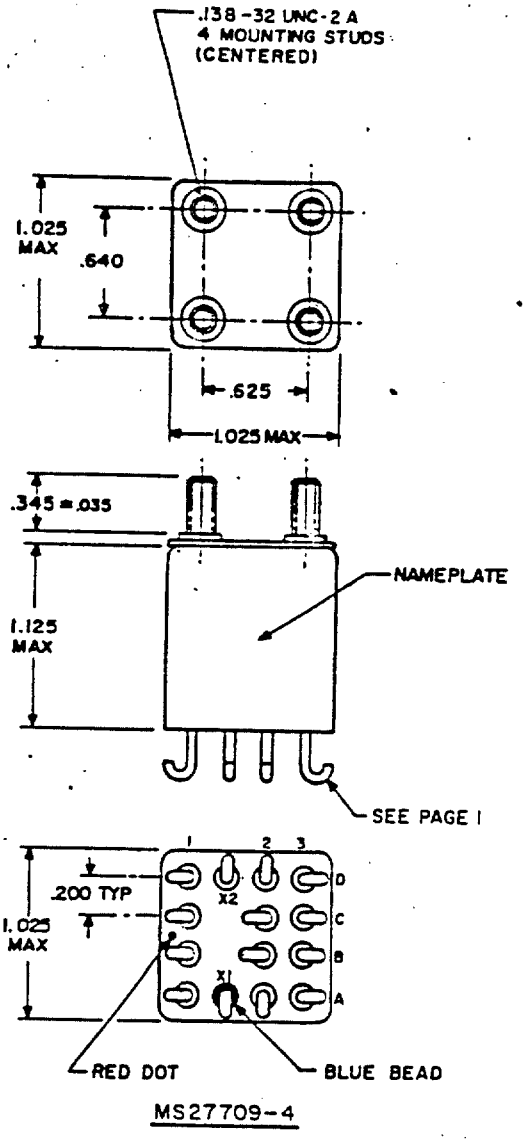
Review activities: Army - EC Navy - Air Force - 99,11

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

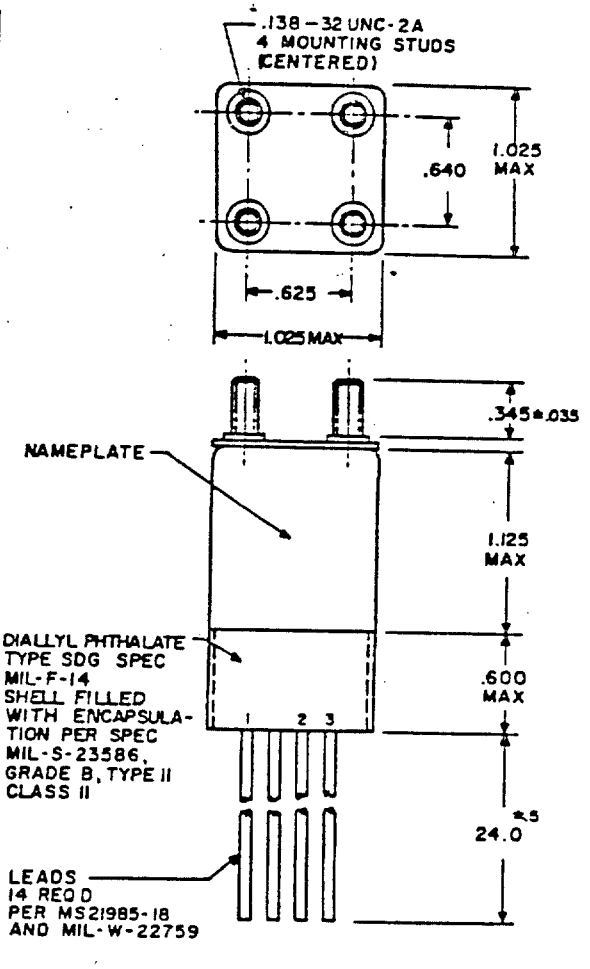
APPROVED 14 July 1967
REVISED (A) 16 MAR 70 (B) 16 DEC 74 (C) 1 AUG 76 (D) 20 FEB 78 (E) 23 JAN 81 (F) 6 JUL 84

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MS27709-4



MS27709-5

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.062	1.57	.230	5.84	.513	13.03	1.359	34.52
.005	.13	.138	3.51	.270	6.86	.600	15.24	1.406	35.71
.020	.51	.144	3.66	.300	7.62	.625	15.88	1.437	36.50
.031	.79	.150	3.81	.312	7.92	.640	16.26	1.750	44.45
.035	.89	.156	3.96	.345	8.76	1.025	26.04	24.0	610.
.050	1.27	.200	5.08	.5	13.	1.125	28.58		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 (.25 mm).
3. Metric equivalents are given for general information only.
4. Terminal numbers need not appear on the relay header, provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. For design feature purposes, this standard takes precedence over procurement documents referenced herein.
6. Referenced documents shall be of the issue in effect on date of invitation for bid.

P.A AF - 85 Other Cust NAVY - AS	International Interest	TITLE (F) RELAY, 10 AMPERES, 4 PDT, TYPE I HERMETICALLY SEALED	MILITARY STANDARD MS27709
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 2 OF 3

DD FORM 1 MAY 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

APPROVED 14 July 1967 REVISED (D) For changes see pages 1 and 2 (E) FOR CHANGES SEE PAGE 1

User activities: Army - Navy - Air Force -

Review activities: Army - EC Navy - Air Force - 99,11

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FED. SUP CLASS 5945

(F)

Operating characteristics.

MS PART NO.	Coil	Coil Data volts	Freq Hz	Res. Ω ±10%	volts	Amp	Normal 1/	High temp test	Cont current test	Drop out voltage 1/	Hold voltage 1/	Time milliseconds - Max 2/ operate	release	Contact Bounce Main NO	Main NC	Aux NO	Aux NC
MS27709-1	X1,X2	28	DC	220	29	.200	18	19.8	22.5	1.5	7.0	15	10	2	2	-	-
-2	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	-	-
-3	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	-	-
-4	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	-	-
-5	X1,X2	28	DC	220	29	.200	18	19.8	22.5	1.5	7.0	15	10	2	2	-	-
-6	X1,X2	115	400	---	122	.04	95	100	105	5	30	20	50	5	5	-	6/

RATED CONTACT LOAD (AMPERES PER POLE) CASE GROUNDED

TYPE OF LOAD	LIFE OPERATING CYCLES X 10³	28 V dc Main NO	Main NC	Aux NO	Aux NC	115 V ac, 1 PHASE Main 400 Hz	Main 60 Hz	Aux 400 Hz	Aux 60 Hz	115/200 V ac, 3 PHASE 3/ Main 400 Hz	Main 60 Hz	Aux 400 Hz	Aux 60 Hz	See Appropriate Notes
RESISTIVE	100	10	10	-	-	10		-	-	10		-	-	
INDUCTIVE				-	-			-	-			-	-	
INDUCTIVE	50	8	8	-	-	8		-	-	8		-	-	
MOTOR	100	4	4	-	-	4		-	-	4		-	-	
LAMP	100	2	2	-	-	2		-	-	2		-	-	
TRANSFER, LOAD				-	-			-	-			-	-	4/
MECHANICAL LIFE REDUCED CURRENT	400	2.5	2.5	-	-	2.5		-	-	2.5		-	-	
MIN CURRENT		APPLICABLE		PER	SPEC									

ENVIRONMENTAL CHARACTERISTICS

TEMPERATURE RANGE ..... −70 °C TO +125 °C
MAX ALTITUDE RATING ..... — 80,000 FT
SHOCK G-LEVEL ..... 50 G
DURATION ..... 10 ms
MAX DURATION CONTACT OPENING ..... 1 ms
VIBRATION - SINUSOIDAL
G - LEVEL ..... 20 G
FREQUENCY RANGE ..... 10 - 2000 Hz
VIBRATION-RANDOM
APPLICABLE SPEC ..... NA
POWER SPECTRAL DENSITY ..... NA
RMS G MIN ..... NA
FREQUENCY RANGE ..... NA
CURVE ..... NA
HIGH SHOCK ..... NA
ACCELERATION ..... 15 G

ELECTRICAL CHARACTERISTICS

INSULATION RESISTANCE, INITIAL ..... 100 MEGOHMS
AFTER LIFE OR ENVIRONMENTAL TESTS .... 50 MEGOHMS

DIELECTRIC STRENGTH (SEA LEVEL)

	Initial	After Life Tests
COIL TO CASE	1000 V RMS	1000 V RMS
AUX CONTACTS	—	—
ALL OTHER POINTS	1500 V RMS	1000 V RMS

5/ DIELECTRIC STRENGTH (ALTITUDE)

	80,000 FT.	
COIL TO CASE	250V RMS	
AUX CONTACTS		
ALL OTHER POINTS	250V RMS	

MAX CONTACT DROP INITIAL ..... .150 VOLTS
AFTER LIFE TEST ..... .175 VOLTS
OVERLOAD CURRENT ..... 40 AMP DC
RUPTURE CURRENT ..... 50 AMP DC
DUTY RATING ..... CONTINUOUS
RFI SPEC
(APPLICABLE TO COIL CIRCUITS OF AC OPERATED RELAYS)

NOTES, APPLICATION

1/ Over temperature range.
2/ With nominal coil voltage.
3/ Absence of value indicates relay is not rated for 3 phase applications.
4/ Transfer load indicates relay suitable for transfer between unsynchronized AC power supplies at rating indicated.
5/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.
6/ At 25°C.

(D) For changes see pages 1 and 2 (E) FOR CHANGES SEE PAGE 1

REVISED
APPROVED 14 July 1967

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Review activities: Army - Navy - EC Air Force - 99,11
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P.A AF - 85
Other Cust NAVY - AS
International Interest

TITLE: RELAY, 10 AMPERES, 4 PDT, TYPE I HERMETICALLY SEALED

MILITARY STANDARD
MS27709

Procurement Specification: MIL-R-6106
SUPERSEDES:

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